

Geometric Control of Quadrotor UAVs Transporting a Cable-Suspended Rigid Body

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Abstract—This paper is focused on tracking control for a rigid body payload that is connected to an arbitrary number of quadrotor unmanned aerial vehicles via rigid links. An intrinsic form of the equations of motion is derived on the nonlinear configuration manifold, and a geometric controller is constructed such that the payload asymptotically follows a given desired trajectory for its position and attitude in the presence of uncertainties. The unique feature is that the coupled dynamics between the rigid body payload, links, and quadrotors are explicitly incorporated into control system design and stability analysis. These are developed in a coordinate-free fashion to avoid singularities and complexities that are associated with local parameterizations. The desirable features of the proposed control system are illustrated by a numerical example.

Index Terms—Aerospace control, Air transportation, Nonlinear control systems.

I. INTRODUCTION

AERIAL transport of payloads by towed cables is common in emergency response, industrial operations, and military missions. Examples of aerial transportation range from search and rescue missions where individuals are lifted from dangerous situations, to the delivery of heavy equipment to the top of a tall building that is difficult to reach by other means.

Transportation of a cable-suspended load has been studied traditionally for human-operated helicopters [1], [2]. Recently, small unmanned aerial vehicles or quadrotors are also considered for load transportation and deployments [3]–[5]. This is to utilize the high thrust-to-weight ratio of quadrotors or multirotor aerial vehicles in autonomous aerial load transportation. However, these are based on simplified dynamics models. For example, the effects of the payload are considered as additional force and torque exerted to quadrotors, instead of considering the dynamic coupling between the payload and the quadrotor, and a precomputed trajectory that minimizes swing motion of the payload is followed, instead of actively controlling the motion of payload and cable [4]. As such, these may not be suitable for agile load transportation where the motion of cable and payload should be actively suppressed online.

Recently, geometric nonlinear control systems are developed for the complete dynamic model of multiple quadrotors

transporting a common payload cooperatively [6]. It is also generalized for a quadrotor with a payload connected by flexible cable that is modeled as a serially connected links, to incorporate the effects of deforming cables [7]. However, in these results, it is assumed that the payload is modeled by a point mass. Such assumption is quite restrictive for practical cases where the size of the payload is comparable to the quadrotors and the length of cables. Also, the effects of modeling errors or unknown disturbances are not considered in the control system design, and therefore, it is impractical to implement those results in experimental settings.

The objective of this paper is to construct a control system for an arbitrary number of quadrotors connected to a rigid body payload via rigid links with explicit consideration on uncertainties. This is challenging in the sense that *dynamically coupled* quadrotors should cooperate safely to transport a rigid body. This is in contrast to the existing results on formation control of decoupled multiagent systems.

In this paper, a coordinate-free form of the equations of motion is derived according to Lagrange mechanics on a nonlinear manifold, and a geometric proportional–integral–derivative (PID) type control system is designed such that the rigid body payload asymptotically follows a given desired trajectory of both the payload position and attitude. The unique property of the proposed control system is that the nontrivial coupling effects between the dynamics of payload, cable, and multiple quadrotors are explicitly incorporated into control system design, without any simplifying assumption. Compared with the preliminary work in [8], this paper adopts nonlinear PID controls and finite-time stability such that the tracking errors asymptotically converge to zero in the presence of uncertainties.

Another distinct feature is that the equations of motion and the control systems are developed directly on the nonlinear configuration manifold intrinsically. Therefore, singularities of local parameterization are completely avoided to generate agile maneuvers of the payload in a uniform way.

In short, the proposed control system is particularly useful for rapid and safe payload transportation in complex terrain, where the position and attitude of the payload should be controlled concurrently in a fast manner to avoid collision with obstacles. Most of the existing control systems of aerial load transportation suffer from limited agility as they are based on reactive assumptions that ignore important dynamic characteristics of aerial load transportation. The proposed control system explicitly integrates the comprehensive dynamic characteristics to achieve extreme maneuverability in aerial load transportation.

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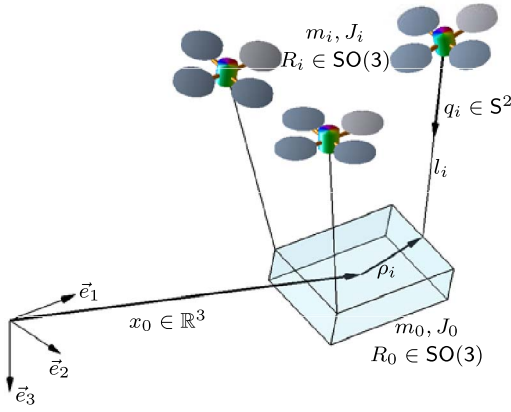


Fig. 1. Dynamics model: n quadrotors connect to a rigid body m_0 via massless links l_i . The configuration manifold is $\mathbb{R}^3 \times \text{SO}(3) \times (\mathbb{S}^2 \times \text{SO}(3))^n$.

II. PROBLEM FORMULATION

Consider n quadrotor UAVs that are connected to a payload that is modeled as a rigid body via massless links (see Fig. 1). Throughout this paper, the variables related to the payload is denoted by the subscript 0, and the variables for the i th quadrotor are denoted by the subscript i , which is assumed to be an element of $\mathcal{I} = \{1, \dots, n\}$ if not specified. We choose an inertial reference frame $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ and body-fixed frames $\{\vec{b}_{j1}, \vec{b}_{j2}, \vec{b}_{j3}\}$ for $0 \leq j \leq n$ as follows. For the inertial frame, the third axis $\vec{e}_3$ points downward along the gravity and the other axes are chosen to form an orthonormal frame. The origin of the j th body-fixed frame is located at the center of mass of the payload for $j = 0$ and at the mass center, the quadrotor for $1 \leq j \leq n$. The third body-fixed axis $\vec{b}_{i3}$ is normal to the plane defined by the centers of rotors, and it points downward.

The location of the mass center of the payload is denoted by $x_0 \in \mathbb{R}^3$, and its attitude is given by $R_0 \in \text{SO}(3)$, where the special orthogonal group is defined by $\text{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \det[R] = 1\}$. Let $\rho_i \in \mathbb{R}^3$ be the point on the payload where the i th link is attached, and it is represented with respect to the zeroth body-fixed frame. The other end of the link is attached to the mass center of the i th quadrotor. The direction of the link from the mass center of the i th quadrotor toward the payload is defined by the unit-vector $q_i \in \mathbb{S}^2$, where $\mathbb{S}^2 = \{q \in \mathbb{R}^3 \mid \|q\| = 1\}$, and the length of the i th link is denoted by $l_i \in \mathbb{R}$. Let $x_i \in \mathbb{R}^3$ be the location of the mass center of the i th quadrotor with respect to the inertial frame. As the link is assumed to be rigid, we have $x_i = x_0 + R_0 \rho_i - l_i q_i$. The attitude of the i th quadrotor is defined by $R_i \in \text{SO}(3)$, which represents the linear transformation of the representation of a vector from the i th body-fixed frame to the inertial frame.

In summary, the configuration of the presented system is described by the position x_0 and the attitude R_0 of the payload, the direction q_i of the links, and the attitudes R_i of the quadrotors. The corresponding configuration manifold of this system is $\mathbb{R}^3 \times \text{SO}(3) \times (\mathbb{S}^2 \times \text{SO}(3))^n$.

The mass and the inertia matrix of the payload are denoted by $m_0 \in \mathbb{R}$ and $J_0 \in \mathbb{R}^{3 \times 3}$, respectively. The dynamic model of each quadrotor is identical to [9]. The mass and the inertia

matrix of the i th quadrotor are denoted by $m_i \in \mathbb{R}$ and $J_i \in \mathbb{R}^{3 \times 3}$, respectively. The i th quadrotor can generate a thrust $-f_i R_i e_3 \in \mathbb{R}^3$ with respect to the inertial frame, where $f_i \in \mathbb{R}$ is the total thrust magnitude and $e_3 = [0, 0, 1]^T \in \mathbb{R}^3$. It also generates a moment $M_i \in \mathbb{R}^3$ with respect to its body-fixed frame. The control input of this system corresponds to $\{f_i, M_i\}_{1 \leq i \leq n}$.

Disturbances are modeled as follows. Let $\Delta_{x_0}, \Delta_{R_0} \in \mathbb{R}^3$ be arbitrary, but fixed disturbance force and moment acting on the mass center of the payload, respectively. The disturbance force and moment exerted on the i th quadrotor are denoted by $\Delta_{x_i}, \Delta_{R_i} \in \mathbb{R}^3$, respectively. The disturbance forces are represented with respect to the inertial frame, and the disturbance moments are represented with respect to the corresponding body-fixed frame.

A. Equations of Motion

The kinematic equations are given by

$$\dot{q}_i = \omega_i \times q_i = \hat{\omega}_i q_i \quad (1)$$

$$\dot{R}_0 = R_0 \hat{\Omega}_0, \quad \dot{R}_i = R_i \hat{\Omega}_i \quad (2)$$

where $\omega_i \in \mathbb{R}^3$ is the angular velocity of the i th link, satisfying $q_i \cdot \omega_i = 0$, and Ω_0 and $\Omega_i \in \mathbb{R}^3$ are the angular velocities of the payload and the i th quadrotor expressed with respect to its body-fixed frame, respectively. The hat map $\hat{\cdot} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ is defined by the condition that $\hat{x}y = x \times y$ for all $x, y \in \mathbb{R}^3$, and the inverse of the hat map is denoted by the vee map $\vee : \mathfrak{so}(3) \rightarrow \mathbb{R}^3$, where $\mathfrak{so}(3)$ denotes the set of 3×3 skew-symmetric matrices, i.e., $\mathfrak{so}(3) = \{S \in \mathbb{R}^{3 \times 3} \mid S^T = -S\}$, and it corresponds to the Lie algebra of $\text{SO}(3)$.

We derive equations of motion according to Lagrangian mechanics. The velocity of the i th quadrotor is given by $\dot{x}_i = \dot{x}_0 + \dot{R}_0 \rho_i - l_i \dot{q}_i$. The kinetic energy of the system is given by

$$\begin{aligned} \mathcal{T} = & \frac{1}{2} m_0 \|\dot{x}_0\|^2 + \frac{1}{2} \Omega_0 \cdot J_0 \Omega_0 \\ & + \sum_{i=1}^n \frac{1}{2} m_i \|\dot{x}_0 + \dot{R}_0 \rho_i - l_i \dot{q}_i\|^2 + \frac{1}{2} \Omega_i \cdot J_i \Omega_i. \end{aligned} \quad (3)$$

The gravitational potential energy is given by

$$\mathcal{U} = -m_0 g e_3 \cdot x_0 - \sum_{i=1}^n m_i g e_3 \cdot (x_0 + R_0 \rho_i - l_i q_i) \quad (4)$$

where the unit-vector e_3 points downward along the gravitational acceleration as shown at Fig. 1. The corresponding Lagrangian of the system is $\mathcal{L} = \mathcal{T} - \mathcal{U}$.

Coordinate-free form of Lagrangian mechanics on the two-sphere $\mathbb{S}^2$ and the special orthogonal group $\text{SO}(3)$ for various multibody systems has been studied in [11] and [12]. By following the same approach, the equations of motion can be obtained from Hamilton's variational principle as follows (see [13] for details):

$$\begin{aligned} & M_q(\ddot{x}_0 - g e_3) - \sum_{i=1}^n m_i q_i q_i^T R_0 \hat{\rho}_i \dot{\Omega}_0 \\ & = \Delta_{x_0} + \sum_{i=1}^n u_i^\parallel + \Delta_{x_i} - m_i l_i \|\omega_i\|^2 q_i - m_i q_i q_i^T R_0 \hat{\Omega}_0^2 \rho_i \end{aligned} \quad (5)$$

$$\begin{aligned}
& \times \left(J_0 - \sum_{i=1}^n m_i \hat{\rho}_i R_0^T q_i q_i^T R_0 \hat{\rho}_i \right) \dot{\Omega}_0 \\
& + \sum_{i=1}^n m_i \hat{\rho}_i R_0^T q_i q_i^T (\ddot{x}_0 - g e_3) + \hat{\Omega}_0 J_0 \Omega_0 = \Delta_{R_0} \\
& + \sum_{i=1}^n \hat{\rho}_i R_0^T (u_i^\parallel + \Delta_{x_i}^\parallel - m_i l_i \|\omega_i\|^2 q_i - m_i q_i q_i^T R_0 \hat{\Omega}_0^2 \rho_i)
\end{aligned} \tag{6}$$

$$\begin{aligned}
\dot{\omega}_i &= \frac{1}{l_i} \hat{q}_i (\ddot{x}_0 - g e_3 - R_0 \hat{\rho}_i \dot{\Omega}_0 + R_0 \hat{\Omega}_0^2 \rho_i) \\
&\quad - \frac{1}{m_i l_i} \hat{q}_i (u_i^\perp + \Delta_{x_i}^\perp)
\end{aligned} \tag{7}$$

$$J_i \dot{\Omega}_i + \Omega_i \times J_i \Omega_i = M_i + \Delta_{R_i} \tag{8}$$

where $M_q = m_y I + \sum_{i=1}^n m_i q_i q_i^T \in \mathbb{R}^{3 \times 3}$, which is symmetric, positive-definite for any q_i .

The vector $u_i \in \mathbb{R}^3$ represents the control force at the i th quadrotor, i.e., $u_i = -f_i R_i e_3$. The vectors $u_i^\parallel$ and $u_i^\perp \in \mathbb{R}^3$ denote the orthogonal projection of u_i along q_i , and the orthogonal projection of u_i to the plane normal to q_i , respectively, that is

$$u_i^\parallel = q_i q_i^T u_i \tag{9}$$

$$u_i^\perp = -\hat{q}_i^2 u_i = (I - q_i q_i^T) u_i. \tag{10}$$

Therefore, $u_i = u_i^\parallel + u_i^\perp$. Throughout this paper, the subscripts $\parallel$ and $\perp$ of a vector denote the component of the vector that is parallel to q_i and the other component of the vector that is perpendicular to q_i . Similarly, the disturbance at the i th quadrotor is decomposed as

$$\Delta_{x_i}^\parallel = q_i q_i^T \Delta_{x_i} \tag{11}$$

$$\Delta_{x_i}^\perp = -\hat{q}_i^2 \Delta_{x_i} = (I - q_i q_i^T) \Delta_{x_i}. \tag{12}$$

B. Tracking Problem

Define a fixed matrix $\mathcal{P} \in \mathbb{R}^{6 \times 3n}$ as

$$\mathcal{P} = \begin{bmatrix} I_{3 \times 3} & \cdots & I_{3 \times 3} \\ \hat{\rho}_1 & \cdots & \hat{\rho}_n \end{bmatrix}. \tag{13}$$

Assume that the links are attached to the payload such that

$$\text{rank}[\mathcal{P}] \geq 6. \tag{14}$$

This is to guarantee that there exist enough degrees of freedom in control inputs for both the translational motion and the rotational maneuver of the payload. The assumption (14) requires that the number of quadrotor is at least three, i.e., $n \geq 3$.

It is also assumed that the bounds of the disturbance forces and moments are available, i.e., for a known positive constant $B_\delta \in \mathbb{R}$

$$\max \{ \|\Delta_{x_0}\|, \|\Delta_{R_0}\|, \left\| \begin{bmatrix} \Delta_{x_1}^T & \cdots & \Delta_{x_n}^T \end{bmatrix}^T \right\|, \|\Delta_{R_i}\| \} < B_\delta. \tag{15}$$

Suppose that the desired trajectories for the position and the attitude of the payload are given as smooth functions of time,

namely $x_{0d}(t) \in \mathbb{R}^3$ and $R_{0d}(t) \in \text{SO}(3)$. From the kinematics equation

$$\dot{R}_{0d}(t) = R_{0d}(t) \hat{\Omega}_{0d}(t)$$

where $\Omega_{0d}(t) \in \mathbb{R}^3$ corresponds to the desired angular velocity of the payload. It is assumed that the velocity and the acceleration of the desired trajectories are bounded by known constants.

We wish to design a control input of each quadrotor $\{f_i, M_i\}_{1 \leq i \leq n}$ such that the tracking errors asymptotically converge to zero along the solution of the controlled dynamics.

III. CONTROL SYSTEM DESIGN FOR SIMPLIFIED DYNAMIC MODEL

In this section, we consider a simplified dynamic model where the attitude dynamics of each quadrotor is ignored, and we design a control input by assuming that the thrust at each quadrotor, namely u_i can be arbitrarily chosen. It corresponds to the case where every quadrotor is replaced by a fully actuated aerial vehicle that can generate a thrust along any direction arbitrarily.

In the simplified dynamic model given by (5)–(7), the dynamics of the payload are affected by the parallel components $u_i^\parallel$ of the thrusts, and the dynamics of the links are directly affected by the normal components $u_i^\perp$ of the thrusts. This structure motivates the following control system design procedure: first, the parallel components $u_i^\parallel$ are chosen such that the payload follows the desired position and attitude trajectory while yielding the desired direction of each link, namely $q_{id} \in \mathbb{S}^2$; next, the normal components $u_i^\perp$ are designed such that the actual direction of the links q_i follows the desired direction q_{id} .

A. Design of Parallel Components

Let $a_i \in \mathbb{R}^3$ be the acceleration of the point on the payload where the i th link is attached, measured relative to the gravity

$$a_i = \ddot{x}_0 - g e_3 + R_0 \hat{\Omega}_0^2 \rho_i - R_0 \hat{\rho}_i \dot{\Omega}_0. \tag{16}$$

The parallel component of the control input is chosen as

$$u_i^\parallel = \mu_i + m_i l_i \|\omega_i\|^2 q_i + m_i q_i q_i^T a_i \tag{17}$$

where $\mu_i \in \mathbb{R}^3$ is a virtual control input that is designed later, with a constraint that μ_i is parallel to q_i . Note that the expression of $u_i^\parallel$ is guaranteed to be parallel to q_i due to the projection operator $q_i q_i^T$ at the last term of the right-hand side of the above-mentioned expression.

The motivation for the proposed parallel components becomes clear if (17) is substituted into (5) and (6) and rearranged to obtain

$$m_0(\ddot{x}_0 - g e_3) = \Delta_{x_0} + \sum_{i=1}^n (\mu_i + \Delta_{x_i}^\parallel) \tag{18}$$

$$J_0 \dot{\Omega}_0 + \hat{\Omega}_0 J_0 \Omega_0 = \Delta_{R_0} + \sum_{i=1}^n \hat{\rho}_i R_0^T (\mu_i + \Delta_{x_i}^\parallel). \tag{19}$$

Therefore, considering a free-body diagram of the payload, the virtual control input μ_i corresponds to the force exerted to the payload by the i -link, or the tension of the i th link in the absence of disturbances.

Next, we determine the virtual control input μ_i . Any control scheme developed for the translational and rotational dynamics of a rigid body can be applied to (18) and (19). Here, we consider a PID control scheme to handle the effects of the disturbances. As in [14], define position, attitude, and angular velocity tracking error vectors $e_{x_0}, e_{R_0}, e_{\Omega_0} \in \mathbb{R}^3$ for the payload as

$$\begin{aligned} e_{x_0} &= x_0 - x_{0d} \\ e_{R_0} &= \frac{1}{2}(R_{0d}^T R_0 - R_0^T R_{0d})^\vee \\ e_{\Omega_0} &= \Omega_0 - R_0^T R_{0d} \Omega_{0d}. \end{aligned}$$

The desired resultant control force $F_d \in \mathbb{R}^3$ and moment $M_d \in \mathbb{R}^3$ acting on the payload are given in terms of these error variables as

$$\begin{aligned} F_d &= m_0(-k_{x_0}e_{x_0} - k_{\dot{x}_0}\dot{e}_{x_0} + \ddot{x}_{0d} - ge_3) \\ &\quad - \bar{\Delta}_{x_0} - \sum_{i=1}^n \bar{\Delta}_{x_i}^\parallel \end{aligned} \quad (20)$$

$$\begin{aligned} M_d &= -k_{R_0}e_{R_0} - k_{\Omega_0}e_{\Omega_0} + (R_0^T R_{0d} \Omega_{0d})^\wedge J_0 R_0^T R_{0d} \Omega_{0d} \\ &\quad + J_0 R_0^T R_{0d} \dot{\Omega}_{0d} - \bar{\Delta}_{R_0} - \sum_{i=1}^n \hat{\rho}_i R_0 \bar{\Delta}_{x_i}^\parallel \end{aligned} \quad (21)$$

for positive constants $k_{x_0}, k_{\dot{x}_0}, k_{R_0}, k_{\Omega_0} \in \mathbb{R}$. Here, the estimates of $\Delta_{x_0}, \Delta_{x_i}, \Delta_{R_0}$ are denoted by $\bar{\Delta}_{x_0}, \bar{\Delta}_{x_i}, \bar{\Delta}_{R_0} \in \mathbb{R}^3$, and the orthogonal projection of $\bar{\Delta}_{x_i}$ along q_i is denoted by $\bar{\Delta}_{x_i}^\parallel = q_i q_i^T \bar{\Delta}_{x_i} \in \mathbb{R}^3$. Integral control laws to update the estimates of disturbances are introduced later in Section III-C.

These are the ideal resultant force and moment to achieve the control objectives. One may try to choose the virtual control input μ_i by making the expressions in the right-hand sides of (18) and (19), namely $\sum_i \mu_i$ and $\sum_i \hat{\rho}_i R_0^T \mu_i$, become identical to F_d and M_d , respectively. But, this is not valid in general, as each μ_i is constrained to be parallel to q_i . Instead, we choose the desired value of μ_i , without any constraint, such that

$$\sum_{i=1}^n \mu_{id} = F_d, \quad \sum_{i=1}^n \hat{\rho}_i R_0^T \mu_{id} = M_d \quad (22)$$

or equivalently, using the matrix $\mathcal{P}$ defined at (13)

$$\mathcal{P} \begin{bmatrix} R_0^T \mu_{1d} \\ \vdots \\ R_0^T \mu_{nd} \end{bmatrix} = \begin{bmatrix} R_0^T F_d \\ M_d \end{bmatrix}.$$

From the assumption stated in (14), there exists at least one solution to the above-mentioned matrix equation for any F_d, M_d . Here, we find the minimum-norm solution given by

$$\begin{bmatrix} \mu_{1d} \\ \vdots \\ \mu_{nd} \end{bmatrix} = \text{diag}[R_0, \dots, R_0] \mathcal{P}^T (\mathcal{P} \mathcal{P}^T)^{-1} \begin{bmatrix} R_0^T F_d \\ M_d \end{bmatrix}. \quad (23)$$

The virtual control input μ_i is selected as

$$\mu_i = (\mu_{id} \cdot q_i) q_i = q_i q_i^T \mu_{id} \quad (24)$$

and the desired direction of each link, namely $q_{id} \in \mathbb{S}^2$ is defined as

$$q_{id} = -\frac{\mu_{id}}{\|\mu_{id}\|}. \quad (25)$$

It is straightforward to verify that when $q_i = q_{id}$, the resultant force and moment acting on the payload become identical to their desired values.

B. Design of Normal Components

Substituting (16) into (7) and using (12), the equation of motion for the i -link is given by

$$\dot{\omega}_i = \frac{1}{l_i} \hat{q}_i a_i - \frac{1}{m_i l_i} \hat{q}_i (u_i^\perp + \Delta_{x_i}^\perp). \quad (26)$$

Here, the normal component of the control input $u_i^\perp$ is chosen such that $q_i \rightarrow q_{id}$ as $t \rightarrow \infty$. Control systems for the unit-vectors on the two-sphere have been studied in [16]. In this paper, we adopt the results of [16], and we augment it with an integral term to handle the disturbance $\Delta_{x_i}^\perp$.

For the given desired direction of each link, its desired angular velocity is obtained from the kinematics equation as

$$\omega_{id} = q_{id} \times \dot{q}_{id}.$$

Define the direction and the angular velocity tracking error vectors for the i th link, namely $e_{q_i}, e_{\omega_i} \in \mathbb{R}^3$ as

$$e_{q_i} = q_{id} \times q_i, \quad e_{\omega_i} = \omega_i + \hat{q}_i^2 \omega_{id}.$$

For positive constants $k_q, k_\omega \in \mathbb{R}$, the normal component of the control input is chosen as

$$\begin{aligned} u_i^\perp &= m_i l_i \hat{q}_i \{ -k_q e_{q_i} - k_\omega e_{\omega_i} - (q_i \cdot \omega_{id}) \hat{q}_i - \hat{q}_i^2 \dot{\omega}_d \} \\ &\quad - m_i \hat{q}_i^2 a_i - \bar{\Delta}_{x_i}^\perp \end{aligned} \quad (27)$$

where $\bar{\Delta}_{x_i}^\perp \in \mathbb{R}^3$ corresponds to the component of the estimate $\bar{\Delta}_{x_i}$ that is projected onto the plane normal to q_i , i.e., $\bar{\Delta}_{x_i}^\perp = -\hat{q}_i^2 \bar{\Delta}_{x_i}$.

Note that the expression of $u_i^\perp$ is perpendicular to q_i by definition. Substituting (27) into (26), and rearranging by the facts that the matrix $-\hat{q}_i^2$ corresponds to the orthogonal projection to the plane normal to q_i and $\hat{q}_i^3 = -\hat{q}_i$, we obtain

$$\dot{\omega}_i = -k_q e_{q_i} - k_\omega e_{\omega_i} - (q_i \cdot \omega_{id}) \hat{q}_i - \hat{q}_i^2 \dot{\omega}_d - \frac{1}{m_i l_i} \hat{q}_i \tilde{\Delta}_{x_i}^\perp \quad (28)$$

where the estimation error is defined as $\tilde{\Delta}_{x_i}^\perp = \Delta_{x_i}^\perp - \bar{\Delta}_{x_i}^\perp \in \mathbb{R}^3$.

In short, the control force for the simplified dynamic model is

$$u_i = u_i^\parallel + u_i^\perp. \quad (29)$$

C. Design of Integral Control Term

The estimates are constructed by the following integral actions:

$$\dot{\bar{\Delta}}_{x_0} = \frac{h_{x_0}}{m_0}(\dot{e}_{x_0} + c_x e_{x_0}) \quad (30)$$

$$\dot{\bar{\Delta}}_{R_0} = h_{R_0}(e_{\Omega_0} + c_R e_{R_0}) \quad (31)$$

$$\begin{aligned} \dot{\bar{\Delta}}_{x_i} = & h_{x_i} q_i q_i^T \left\{ \frac{1}{m_0}(\dot{e}_{x_0} + c_x e_{x_0}) - R_0 \hat{\rho}_i (e_{\Omega_0} + c_R e_{R_0}) \right\} \\ & + \frac{h_{x_i}}{m_i l_i} \hat{q}_i (e_{\omega_i} + c_q e_{q_i}) \end{aligned} \quad (32)$$

for positive constants $c_x, c_R, c_q \in \mathbb{R}$ and integral gains $h_{x_0}, h_{R_0}, h_{x_i} \in \mathbb{R}$.

Proposition 1: Consider the simplified dynamic model defined by (5)–(7). For given tracking commands x_{0d}, R_{0d} , a control input is designed as (29)–(32). Then, there exist the values of controller gains and controller parameters such that the following properties are satisfied.

- 1) The zero equilibrium of tracking errors $(e_{x_0}, \dot{e}_{x_0}, e_{R_0}, e_{\Omega_0}, e_{q_i}, e_{\omega_i})$ and the estimation errors $(\bar{\Delta}_{x_0}, \bar{\Delta}_{R_0}, \bar{\Delta}_{x_i})$ is stable in the sense of Lyapunov.
- 2) The tracking errors asymptotically coverage to zero.
- 3) The estimation errors are uniformly bounded.

Proof: See Appendix A $\square$

The proposed control system guarantees stability in the sense of Lyapunov and asymptotic convergence of tracking errors variables, but as the convergence of estimation error is not guaranteed, asymptotic stability is not achieved. However, in the absence of the disturbances, we can achieve stronger exponential stability by eliminating the integral terms of the proposed control system via setting $\bar{\Delta}_{x_0} = \bar{\Delta}_{R_0} = \bar{\Delta}_{x_i} = 0$ and $h_{x_0}, h_{R_0}, h_{x_i} = 0$ [8].

IV. CONTROL SYSTEM DESIGN FOR FULL DYNAMIC MODEL

The control system designed in Section III is based on a simplifying assumption that each quadrotor can generate a thrust along any arbitrary direction instantaneously. However, the dynamics of quadrotor is underactuated, since the direction of the total thrust is always parallel to its third body-fixed axis, while the magnitude of the total thrust can be arbitrarily changed. Whereas the rotational attitude dynamics is fully actuated by the control moment M_i .

Based on these observations, the attitude of each quadrotor is controlled such that the third body-fixed axis becomes parallel to the direction of the ideal control force u_i designed in Section III within a finite time. More explicitly, the desired direction of the third body-fixed axis of the i th quadrotor, namely $b_{3_i} \in \mathbb{S}^2$ is

$$b_{3_i} = -\frac{u_i}{\|u_i\|}. \quad (33)$$

This provides 2-D constraint on the 3-D desired attitude of each quadrotor, and there remains one degree of freedom. To resolve it, the desired direction of the first body-fixed axis $b_{1_i}(t) \in \mathbb{S}^2$ is introduced as a smooth function of time. Due to the fact that the first body-fixed axis is normal to the

third body-fixed axis, it is impossible to follow an arbitrary command $b_{1_i}(t)$ exactly. Instead, its projection onto the plane normal to b_{3_i} is followed, and the desired direction of the second body-fixed axis is chosen to constitute an orthonormal frame [9]. From these, the desired attitude of the i th quadrotor is given by

$$R_{i_c} = \left[-\frac{(\hat{b}_{3_i})^2 b_{1_i}}{\|\hat{b}_{3_i}\|^2 b_{1_i}}, \frac{\hat{b}_{3_i} b_{1_i}}{\|\hat{b}_{3_i}\| b_{1_i}}, b_{3_i} \right] \in \text{SO}(3).$$

The desired angular velocity is obtained from the attitude kinematics equation, i.e., $\Omega_{i_c} = (R_{i_c}^T \dot{R}_{i_c})^\vee \in \mathbb{R}^3$.

In the prior work described in [8], the stability of the combined full dynamic model is achieved by exploiting exponential convergence of the attitude errors and singular perturbation [17]. However, we cannot follow such approach in this paper, as the presented PID control system guarantees only asymptotical convergence of the tracking errors due to the disturbances. Here, we design the attitude controller of each quadrotor such that R_i becomes equal to R_{i_c} within a finite time via finite-time stability theory [18]–[20].

Define the tracking error vectors $e_{R_i}, e_{\Omega_i} \in \mathbb{R}^3$ for the attitude and the angular velocity of the i th quadrotor as

$$e_{R_i} = \frac{1}{2}(R_{i_c}^T R_i - R_i^T R_{i_c})^\vee, \quad e_{\Omega_i} = \Omega_i - R_i^T R_{i_c} \Omega_{i_c}.$$

The time-derivative of e_{R_i} can be written as [9]

$$\dot{e}_{R_i} = \frac{1}{2}(\text{tr}[R_i^T R_{i_c}]I - R_i^T R_{i_c})e_{\Omega_i} \triangleq E(R_i, R_{i_c})e_{\Omega_i}. \quad (34)$$

For $0 < r < 1$, define $S: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ as

$$S(r, y) = [|y_1|^r \text{sgn}(y_1), |y_2|^r \text{sgn}(y_2), |y_3|^r \text{sgn}(y_3)]^T$$

where $y = [y_1, y_2, y_3]^T \in \mathbb{R}^3$, and $\text{sgn}(\cdot)$ denotes the sign function. For positive constants k_R, l_R , the terminal sliding surface $s_i \in \mathbb{R}^3$ is

$$s_i = e_{\Omega_i} + k_R e_{R_i} + l_R S(r, e_{R_i}). \quad (35)$$

We can show that when confined to the surface of $s_i \equiv 0$, the tracking errors become zero in a finite time. To reach the sliding surface, for positive constants k_s, l_s , the control moment is designed as

$$\begin{aligned} M_i = & -k_s s_i - l_s S(r, s_i) + \Omega_i \times J_i \Omega_i \\ & - (k_R J_i + l_s r J_i \text{diag}_j[|e_{R_{ij}}|^{r-1}])E(R_i, R_{i_c})e_{\Omega_i} \\ & - J_i(\hat{\Omega}_i R_i^T R_{i_c} \Omega_{i_c} - R_i^T R_{i_c} \dot{\Omega}_{i_c}). \end{aligned} \quad (36)$$

The thrust magnitude is chosen as

$$f_i = -u_i \cdot R_i e_3 \quad (37)$$

which yields that the thrust of each quadrotor becomes equal to its desired value u_i when $R_i = R_{i_c}$.

Stability of the corresponding controlled systems for the full dynamic model can be shown by using the fact that the full dynamic model becomes exactly the same as the simplified dynamic model within a finite time.

Proposition 2: Consider the full dynamic model defined by (5)–(8). For given tracking commands x_{0d}, R_{0d} and the desired direction of the first body-fixed axis b_{1_i} , control inputs

for quadrotors are designed as (36) and (37). Then, there exist controller parameters such that the tracking error variables $(e_{x_0}, \dot{e}_{x_0}, e_{R_0}, \dot{e}_{R_0}, e_{q_i}, \dot{e}_{q_i}, e_{\omega_i}, \dot{e}_{\omega_i})$ asymptotically converge to zero, and the estimation errors are uniformly bounded.

Proof: See Appendix B. $\square$

In contrast to the existing results in aerial transportation of a cable-suspended load, it does not rely on any simplifying assumption that ignores the coupling between payload, cable, and quadrotors. Also, the presented global formulation on the nonlinear configuration manifold avoids singularities and complexities that are inherently associated with local coordinates. As such, the presented control system is particularly useful for agile load transportation involving combined translational and rotational maneuvers of the payload.

V. NUMERICAL EXAMPLE

We consider a numerical example where three quadrotors ($n = 3$) transport a rectangular box along a figure-eight curve. More explicitly, the mass of the payload is $m_0 = 1.5$ kg, and its length, width, and height are 1.0 m, 0.8 m, and 0.2 m, respectively. Throughout this section, all of the units are in kg, m, and s, if not specified. Mass properties of three quadrotors are identical, and they are given by $m_i = 0.755$, $J_i = \text{diag}[0.0820, 0.0845, 0.1377]$. The length of cable is $l_i = 1$ m, and they are attached to the following points of the payload: $\rho_1 = [0.5, 0, -0.1]^T$, $\rho_2 = [-0.5, 0.4, -0.1]^T$, $\rho_3 = [-0.5, -0.4, -0.1]^T$ (see Fig. 1).

The desired trajectory of the payload is chosen as

$$x_{0d}(t) = [1.2 \sin(0.2\pi t), 4.2 \cos(0.1\pi t), -0.5]^T.$$

The desired attitude of the payload is chosen such that its first axis is tangent to the desired path, and the third axis is parallel to the direction of gravity, it is given by

$$R_{0d}(t) = \begin{bmatrix} \frac{\dot{x}_{0d}}{\|\dot{x}_{0d}\|} & \frac{\hat{e}_3 \dot{x}_{0d}}{\|\hat{e}_3 \dot{x}_{0d}\|} & e_3 \end{bmatrix}.$$

Initial conditions are chosen as $x_0(0) = [1, 4.8, 0]^T$, $v_0(0) = 0_{3 \times 1}$, $q_i(0) = e_3$, $\omega_i(0) = 0_{3 \times 1}$, $R_i(0) = I_{3 \times 3}$, $\Omega_i(0) = 0_{3 \times 1}$.

While the proposed control system relies on the assumption that the uncertain terms are fixed, the following time-varying disturbances are considered in the numerical simulations:

$$\begin{aligned} \Delta_{x_0} &= [1 + 0.1 \sin 2t, 3 + 0.3 \cos t, -2.5 - 0.2 \sin 3t]^T \\ \Delta_{R_0} &= [-0.5 - 0.07 \sin t, 0.1 \\ &\quad + 0.02 \cos 2t, -1.5 - 0.22 \sin 1.5t]^T \\ \Delta_{x_i} &= [0.5 + 0.08 \sin 4t, \\ &\quad -0.2 - 0.16 \cos 2t, 0.3 + 0.1 \sin t]^T \\ \Delta_{R_i} &= [0.2 - 0.05 \cos 3t, 0.3 \\ &\quad -0.1 \sin 2t, -0.7 - 0.1 \sin 2t]^T. \end{aligned}$$

The corresponding simulation results are shown at Figs. 2 and 3. Fig. 2 shows tracking errors for the position and the attitude of the payload, as well as tension and control inputs. The corresponding simulation results are shown in Figs. 2 and 3. Fig. 2 shows tracking errors for the position and the attitude of the payload, as well as tension and control inputs. The green curves and the blue curves correspond to

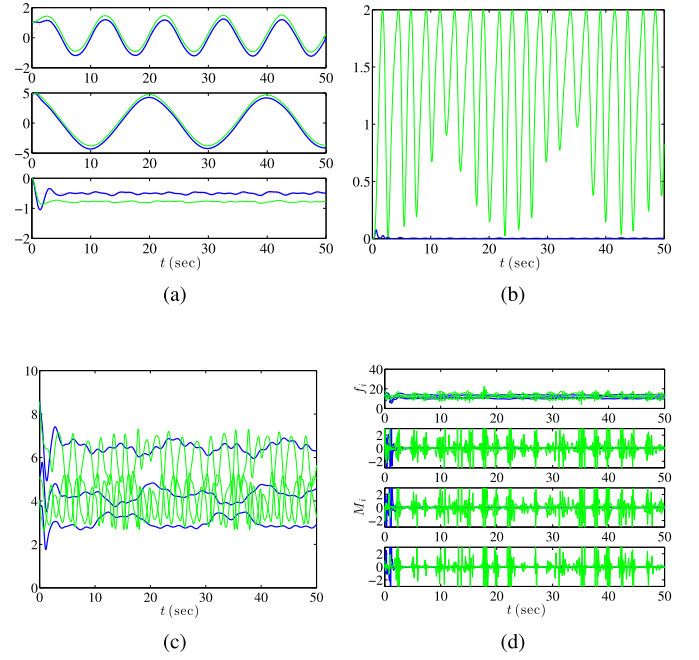


Fig. 2. Simulation results for the payload tracking errors and control inputs (green curves: without the integral term; blue curves: with the proposed integral term). (a) Position of payload (red: x_0, x_{0d}). (b) Attitude tracking error of payload $\Psi_0 = 1/2 \|R_0 - R_{0d}\|^2$. (c) Tension at links. (d) Control input for quadrotors f_i, M_i .

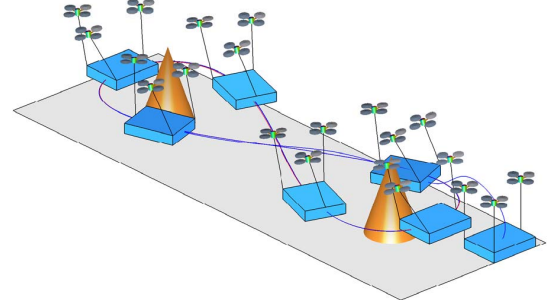


Fig. 3. Snapshots of controlled maneuver with the integral term (red curves: desired trajectory, blue curves: actual trajectory).

the simulations results of the control systems without integral terms [8], and the results of the proposed control system with integral terms, respectively. Without the integral term (green curves), the uncertainties cause substantial errors for the payload in following its desired trajectories, and in particular the attitude errors reach the maximum values of 180° repeatedly. Furthermore, there exist high-frequency oscillations in the tension of the links and the control inputs of quadrotors. The proposed integral terms (blue curves), successfully suppress those undesired behaviors caused by the uncertainties, and they exhibit excellent tracking performances of the proposed control system. Fig. 3 illustrates the controlled maneuvers of the payload and quadrotors with the integral terms, following a figure-eight curve around two obstacles represented by cones.

VI. CONCLUSION

This paper presents a geometric control system for an arbitrary number of quadrotors that transport a cable-suspended

rigid body. It is guaranteed that the payload asymptotically follows a desired position trajectory and a desired attitude trajectory concurrently in the presence of uncertainties. The main contribution is developing a geometric nonlinear PID control system that incorporate the nontrivial coupling between the payload, links, and quadrotors in an intrinsic fashion. Rigorous stability proof of a nonlinear control system for such comprehensive dynamic model of cooperative aerial load transportation has been unprecedented.

APPENDIX

A. Proof of Proposition 1

1) *Error Dynamics*: From (5) and (24), the dynamics of the position tracking error is given by

$$m_0 \ddot{e}_{x_0} = m_0 (ge_3 - \ddot{x}_{0d}) + \Delta_{x_0} + \sum_{i=1}^n q_i q_i^T \mu_{i_d} + \Delta_{x_i}^{\parallel}.$$

From (22) and (20), this can be rearranged as

$$\begin{aligned} \ddot{e}_{x_0} &= ge_3 - \ddot{x}_{0d} + \frac{1}{m_0} F_d + Y_x + \frac{1}{m_0} \left(\tilde{\Delta}_{x_0} + \sum_{i=1}^n \tilde{\Delta}_{x_i}^{\parallel} \right) \\ &= -k_{x_0} e_{x_0} - k_{\dot{x}_0} \dot{e}_{x_0} + \frac{1}{m_0} \left(\tilde{\Delta}_{x_0} + \sum_{i=1}^n \tilde{\Delta}_{x_i}^{\parallel} \right) + Y_x \end{aligned} \quad (38)$$

where $\tilde{\Delta}_{x_i}^{\parallel} = \Delta_{x_i}^{\parallel} - \bar{\Delta}_{x_i}^{\parallel} \in \mathbb{R}^3$ corresponds to the parallel component of the estimation error. The last term $Y_x \in \mathbb{R}^3$ represents the error caused by the difference between q_i and q_{i_d} , and it is given by

$$Y_x = \frac{1}{m_0} \sum_{i=1}^n (q_i q_i^T - I) \mu_{i_d}.$$

We have $\mu_{i_d} = q_{i_d} q_{i_d}^T \mu_{i_d}$ from (25). Using this, the error term can be written in terms of e_{q_i} as

$$Y_x = -\frac{1}{m_0} \sum_{i=1}^n (q_{i_d}^T \mu_{i_d}) \hat{q}_i e_{q_i}.$$

Using (23), an upper bound of Y_x can be obtained as

$$\|Y_x\| \leq \frac{1}{m_0} \sum_{i=1}^n \|\mu_{i_d}\| \|e_{q_i}\| \leq \sum_{i=1}^n \gamma (\|F_d\| + \|M_d\|) \|e_{q_i}\|$$

where $\gamma = (1/(m_0(\lambda_m[\mathcal{PP}^T]))^{1/2})$. From (20) and (21), this can be further bounded by

$$\begin{aligned} \|Y_x\| &\leq \sum_{i=1}^n \{\beta(k_{x_0} \|e_{x_0}\| + k_{\dot{x}_0} \|\dot{e}_{x_0}\|) \\ &\quad + \gamma(k_{R_0} \|e_{R_0}\| + k_{\Omega_0} \|e_{\Omega_0}\|) + B\|e_{q_i}\| \end{aligned} \quad (39)$$

where $\beta = m_0 \gamma$, and the constant B is determined by the given desired trajectories of the payload and (43), which defines the domain D of the error variables that the presented stability proof is considered. Throughout the remaining parts of the proof, any bound that can be obtained from x_{0d} , R_{0d} or (43) is denoted by B for simplicity. In short, the position tracking error dynamics of the payload can be written as (38), where the error term is bounded by (39).

Similarly, we find the attitude tracking error dynamics for the payload as follows. Using (6), (21), and (24), the time-derivative of $J_0 e_{\Omega_0}$ can be written as

$$\begin{aligned} J_0 \dot{e}_{\Omega_0} &= (J_0 e_{\Omega_0} + d)^{\wedge} e_{\Omega_0} - k_{R_0} e_{R_0} - k_{\Omega_0} e_{\Omega_0} \\ &\quad + \tilde{\Delta}_{R_0} + \sum_{i=1}^n \hat{\rho}_i R_0^T \tilde{\Delta}_{x_i}^{\parallel} + Y_R \end{aligned} \quad (40)$$

where $d = (2J_0 - \text{tr}[J_0]I)R_0^T R_{0d} \Omega_{0d} \in \mathbb{R}^3$ [14] that is bounded, and $\tilde{\Delta}_{R_0} \in \mathbb{R}^3$ denotes the estimation error given by $\tilde{\Delta}_{R_0} = \Delta_{R_0} - \bar{\Delta}_{R_0}$. The error term in the attitude dynamics of the payload, namely $Y_R \in \mathbb{R}^3$ is given by

$$\begin{aligned} Y_R &= \sum_{i=1}^n \hat{\rho}_i R_0^T (q_i q_i^T - I) \mu_{i_d} \\ &= -\sum_{i=1}^n \hat{\rho}_i R_0^T (q_{i_d}^T \mu_{i_d}) \hat{q}_i e_{q_i}. \end{aligned}$$

Similar to (39), an upper bound of Y_R can be obtained as

$$\begin{aligned} \|Y_R\| &\leq \sum_{i=1}^n \{\delta_i(k_{x_0} \|e_{x_0}\| + k_{\dot{x}_0} \|\dot{e}_{x_0}\|) \\ &\quad + \sigma_i(k_{R_0} \|e_{R_0}\| + k_{\Omega_0} \|e_{\Omega_0}\|) + B\|e_{q_i}\| \end{aligned} \quad (41)$$

where $\delta_i = m_0(\|\hat{\rho}_i\|/(\lambda_m[\mathcal{PP}^T]))^{1/2}$, $\sigma_i = \delta_i/m_0 \in \mathbb{R}$.

Next, from (28), the time-derivative of the angular velocity error, projected on to the plane normal to q_i is given as

$$-\hat{q}_i^2 \dot{e}_{\omega_i} = -k_q e_{q_i} - k_{\omega} e_{\omega_i} - \frac{1}{m_i l_i} \hat{q}_i \tilde{\Delta}_{x_i}^{\perp}. \quad (42)$$

In summary, the error dynamics of the simplified dynamic model are given by (38), (40), and (42).

2) *Stability Proof*: Define an attitude configuration error function Ψ_{R_0} for the payload as

$$\Psi_{R_0} = \frac{1}{2} \text{tr}[I - R_{0d}^T R_0]$$

which is positive-definite about $R_0 = R_{0d}$, and $\dot{\Psi}_{R_0} = e_{R_0} \cdot e_{\Omega_0}$ [9], [14]. We also introduce a configuration error function Ψ_{q_i} for each link that is positive-definite about $q_i = q_{i_d}$ as

$$\Psi_{q_i} = 1 - q_i \cdot q_{i_d}.$$

For positive constants $e_{x_{\max}}$, ψ_{R_0} , ψ_{q_i} , $B_{\delta} \in \mathbb{R}$, consider the following open domain:

$$\begin{aligned} D &= \{(e_{x_0}, \dot{e}_{x_0}, e_{R_0}, e_{\Omega_0}, e_{q_i}, e_{\omega_i}, \tilde{\Delta}_{x_0}, \tilde{\Delta}_{R_0}, \tilde{\Delta}_{x_i}) \\ &\quad \in (\mathbb{R}^3)^4 \times (\mathbb{R}^3 \times \mathbb{R}^3)^n \times (\mathbb{R}^3)^2 \times \mathbb{R}^{3n} \mid \\ &\quad \|e_{x_0}\| < e_{x_{\max}}, \Psi_{R_0} < \psi_{R_0} < 1, \Psi_{q_i} < \psi_{q_i} < 1, \\ &\quad \|\tilde{\Delta}_{x_0}\| < B_{\delta}, \|\tilde{\Delta}_{R_0}\| < B_{\delta}, \|\tilde{\Delta}_{x_i}\| < B_{\delta}\}. \end{aligned} \quad (43)$$

In this domain, we have $\|e_{R_0}\| = (\Psi_{R_0}(2 - \Psi_{R_0}))^{1/2} \leq (\psi_{R_0}(2 - \psi_{R_0}))^{1/2} \triangleq \alpha_0 < 1$, and $\|e_{q_i}\| = (\Psi_{q_i}(2 - \Psi_{q_i}))^{1/2} \leq (\psi_{q_i}(2 - \psi_{q_i}))^{1/2} \triangleq \alpha_i < 1$. It is assumed that ψ_{q_i} is sufficiently small such that $n\alpha_i\beta < 1$.

We can show that the configuration error functions are quadratic with respect to the error vectors in the sense that

$$\begin{aligned} \frac{1}{2} \|e_{R_0}\|^2 &\leq \Psi_{R_0} \leq \frac{1}{2 - \psi_{R_0}} \|e_{R_0}\|^2 \\ \frac{1}{2} \|e_{q_i}\|^2 &\leq \Psi_{q_i} \leq \frac{1}{2 - \psi_{q_i}} \|e_{q_i}\|^2 \end{aligned}$$

where the upper bounds are satisfied only in the domain D . Define

$$\begin{aligned}\mathcal{V}_0 = & \frac{1}{2}\|\dot{e}_{x_0}\|^2 + \frac{1}{2}k_{x_0}\|e_{x_0}\|^2 + c_x e_{x_0} \cdot \dot{e}_{x_0} \\ & + \frac{1}{2}e_{\Omega_0} \cdot J_0 \Omega_0 + k_{R_0} \Psi_{R_0} + c_R e_{R_0} \cdot J_0 e_{\Omega_0} \\ & + \sum_{i=1}^n \frac{1}{2}\|e_{\omega_i}\|^2 + k_q \Psi_{q_i} + c_q e_{q_i} \cdot e_{\omega_i}\end{aligned}$$

where c_x, c_R, c_q values are positive constants. This is composed of tracking error variables only, and we define another function for the estimation errors as

$$\mathcal{V}_a = \frac{1}{2h_{x_0}}\|\tilde{\Delta}_{x_0}\|^2 + \frac{1}{2h_{R_0}}\|\tilde{\Delta}_{R_0}\|^2 + \sum_{i=1}^n \frac{1}{2h_{x_i}}\|\tilde{\Delta}_{x_i}\|^2.$$

The Lyapunov function for the complete simplified dynamic model is chosen as $\mathcal{V} = \mathcal{V}_0 + \mathcal{V}_a$.

Let $z_{x_0} = [\|e_{x_0}\|, \|\dot{e}_{x_0}\|]^T$, $z_{R_0} = [\|e_{R_0}\|, \|e_{\Omega_0}\|]^T$, $z_{q_i} = [\|e_{q_i}\|, \|e_{\omega_i}\|]^T \in \mathbb{R}^2$. The first part of the Lyapunov function $\mathcal{V}_0$ satisfies

$$\begin{aligned}z_{x_0}^T \underline{P}_{x_0} z_{x_0} + z_{R_0}^T \underline{P}_{R_0} z_{R_0} + \sum_{i=1}^n z_{q_i}^T \underline{P}_{q_i} z_{q_i} &\leq \mathcal{V}_0 \\ &\leq z_{x_0}^T \bar{P}_{x_0} z_{x_0} + z_{R_0}^T \bar{P}_{R_0} z_{R_0} + \sum_{i=1}^n z_{q_i}^T \bar{P}_{q_i} z_{q_i}\end{aligned}$$

where the matrices $\underline{P}_{x_0}, \underline{P}_{R_0}, \underline{P}_{q_i}, \underline{P}_{x_0}, \underline{P}_{R_0}, \underline{P}_{q_i} \in \mathbb{R}^{2 \times 2}$ are given by

$$\begin{aligned}\underline{P}_{x_0} &= \frac{1}{2} \begin{bmatrix} k_{x_0} & -c_x \\ -c_x & 1 \end{bmatrix}, \quad \bar{P}_{x_0} = \frac{1}{2} \begin{bmatrix} k_{x_0} & c_x \\ c_x & 1 \end{bmatrix} \\ \underline{P}_{R_0} &= \frac{1}{2} \begin{bmatrix} 2k_{R_0} & -c_R \bar{\lambda} \\ -c_R \bar{\lambda} & \underline{\lambda} \end{bmatrix}, \quad \bar{P}_{R_0} = \frac{1}{2} \begin{bmatrix} \frac{2k_{R_0}}{2 - \psi_{R_0}} & c_R \bar{\lambda} \\ c_R \bar{\lambda} & \bar{\lambda} \end{bmatrix} \\ \underline{P}_{q_i} &= \frac{1}{2} \begin{bmatrix} 2k_q & -c_q \\ -c_q & 1 \end{bmatrix}, \quad \bar{P}_{q_i} = \frac{1}{2} \begin{bmatrix} \frac{2k_q}{2 - \psi_{q_i}} & c_q \\ c_q & 1 \end{bmatrix}\end{aligned}$$

where $\underline{\lambda} = \lambda_m[J_0]$ and $\bar{\lambda} = \lambda_M[J_0]$. If the constants c_x, c_{R_0} , and c_q are sufficiently small, all of the above-mentioned matrices are positive-definite. As the second part of the Lyapunov function $\mathcal{V}_a$ is already given as a quadratic form, it is straightforward to see that the complete Lyapunov function $\mathcal{V}$ is positive-definite and decrescent.

The time-derivative of the Lyapunov function along the error dynamics (38), (40), and (42) is given by

$$\begin{aligned}\dot{\mathcal{V}} = & -(k_{\dot{x}_0} - c_x)\|\dot{e}_{x_0}\|^2 - c_x k_{x_0}\|e_{x_0}\|^2 - c_x k_{\dot{x}_0} e_{x_0} \cdot \dot{e}_{x_0} \\ & + (c_x e_{x_0} + \dot{e}_{x_0}) \cdot Y_x - k_{\Omega_0}\|e_{\Omega_0}\|^2 + c_R \dot{e}_R \cdot J_0 e_{\Omega_0} \\ & - c_R k_{R_0}\|e_{R_0}\|^2 + c_R e_{R_0} \cdot ((J_0 e_{\Omega_0} + d)^\wedge e_{\Omega_0} - k_{\Omega_0} e_{\Omega_0}) \\ & + (e_{\Omega_0} + c_R e_{R_0}) \cdot Y_R \\ & + \sum_{i=1}^n -(k_{\omega} - c_q)\|e_{\omega_i}\|^2 - c_q k_q\|e_{q_i}\|^2 - c_q k_{\omega} e_{q_i} \cdot e_{\omega_i} \\ & + \frac{1}{m_0}(\dot{e}_{x_0} + c_x e_{x_0}) \cdot \left(\tilde{\Delta}_{x_0} + \sum_{i=1}^n \tilde{\Delta}_{x_i} \right)\end{aligned}$$

$$\begin{aligned}& + (e_{\Omega_0} + c_R e_{R_0}) \cdot \left(\tilde{\Delta}_{R_0} + \sum_{i=1}^n \hat{\rho}_i R_0^T \tilde{\Delta}_{x_i} \right) \\ & - \left(\sum_{i=1}^n (e_{\omega_i} + c_q e_{q_i}) \cdot \frac{\hat{q}_i}{m_i l_i} \tilde{\Delta}_{x_i}^\perp \right) - \frac{1}{h_{x_0}} \tilde{\Delta}_{x_0} \cdot \dot{\tilde{\Delta}}_{x_0} \\ & - \frac{1}{h_{R_0}} \tilde{\Delta}_{R_0} \cdot \dot{\tilde{\Delta}}_{R_0} - \sum_{i=1}^n \frac{1}{h_{x_i}} (\tilde{\Delta}_{x_i}^\perp \cdot \dot{\tilde{\Delta}}_{x_i} + \tilde{\Delta}_{x_i} \cdot \dot{\tilde{\Delta}}_{x_i}^\perp)\end{aligned}\quad (44)$$

where the last term has been obtained using the facts that $\tilde{\Delta}_{x_i}^\perp \cdot \dot{\tilde{\Delta}}_{x_i} = \tilde{\Delta}_{x_i} \cdot \dot{\tilde{\Delta}}_{x_i}^\perp = 0$. Substituting (30), (31), and (32) into (44), the expressions at the last four lines of (44) that are dependent of estimation errors vanish.

An upper bound of the remaining expressions of $\dot{\mathcal{V}}$ can be obtained as follows. Since $\|e_{R_0}\| \leq 1$, $\|\dot{e}_{R_0}\| \leq \|e_{\Omega_0}\|$ and $\|d\| \leq B$

$$\begin{aligned}\dot{\mathcal{V}} \leq & -(k_{\dot{x}_0} - c_x)\|\dot{e}_{x_0}\|^2 - c_x k_{x_0}\|e_{x_0}\|^2 - c_x k_{\dot{x}_0} e_{x_0} \cdot \dot{e}_{x_0} \\ & + (c_x e_{x_0} + \dot{e}_{x_0}) \cdot Y_x - (k_{\Omega_0} - 2c_R \bar{\lambda})\|e_{\Omega_0}\|^2 \\ & - c_R k_{R_0}\|e_{R_0}\|^2 + c_R (k_{\Omega_0} + B)\|e_{R_0}\|\|e_{\Omega_0}\| \\ & + (e_{\Omega_0} + c_R e_{R_0}) \cdot Y_R \\ & + \sum_{i=1}^n -(k_{\omega} - c_q)\|e_{\omega_i}\|^2 - c_q k_q\|e_{q_i}\|^2 - c_q k_{\omega} e_{q_i} \cdot e_{\omega_i}.\end{aligned}\quad (45)$$

From (39), an upper bound of the fourth term of the right-hand side is given by

$$\begin{aligned}& \|(c_x e_{x_0} + \dot{e}_{x_0}) \cdot Y_x\| \\ & \leq \sum_{i=1}^n \alpha_i \beta (c_x k_{x_0}\|e_{x_0}\|^2 + c_x k_{\dot{x}_0}\|e_{x_0}\|\|\dot{e}_{x_0}\| + k_{\dot{x}_0}\|\dot{e}_{x_0}\|^2) \\ & \quad + \{c_x B\|e_x\| + (\beta k_{x_0} e_{x_{\max}} + B)\|\dot{e}_{x_0}\|\}\|e_{q_i}\| \\ & \quad + \alpha_i \gamma (c_x\|e_{x_0}\| + \|\dot{e}_{x_0}\|)(k_{R_0}\|e_{R_0}\| + k_{\Omega_0}\|e_{\Omega_0}\|).\end{aligned}\quad (46)$$

Similarly, using (41)

$$\begin{aligned}& \|(c_R e_{R_0} + e_{\Omega_0}) \cdot Y_R\| \\ & \leq \sum_{i=1}^n \alpha_i \sigma_i (c_R k_{R_0}\|e_{R_0}\|^2 + c_R k_{\Omega_0}\|e_{R_0}\|\|e_{\Omega_0}\| + k_{\Omega_0}\|e_{\Omega_0}\|^2) \\ & \quad + \{c_R B\|e_{R_0}\| + (\alpha_0 \sigma_i k_{R_0} + B)\|e_{\Omega_0}\|\}\|e_{q_i}\| \\ & \quad + \alpha_i \delta_i (c_R\|e_{R_0}\| + \|e_{\Omega_0}\|)(k_{x_0}\|e_{x_0}\| + k_{\dot{x}_0}\|\dot{e}_{x_0}\|).\end{aligned}\quad (47)$$

Substituting these into (45) and rearranging

$$\dot{\mathcal{V}} \leq \sum_{i=1}^n -z_i^T W_i z_i \quad (48)$$

where $z_i = [\|z_{x_0}\|, \|z_{R_0}\|, \|z_{q_i}\|]^T \in \mathbb{R}^3$, and the matrix $W_i \in \mathbb{R}^{3 \times 3}$ is defined as

$$W_i = \begin{bmatrix} \lambda_m[W_{x_i}] & -\frac{1}{2}\|W_{xR_i}\| & -\frac{1}{2}\|W_{xq_i}\| \\ -\frac{1}{2}\|W_{xR_i}\| & \lambda_m[W_{R_i}] & -\frac{1}{2}\|W_{Rq_i}\| \\ -\frac{1}{2}\|W_{xq_i}\| & -\frac{1}{2}\|W_{Rq_i}\| & \lambda_m[W_{q_i}] \end{bmatrix} \quad (49)$$

where the submatrices are given by

$$\begin{aligned} W_{x_i} &= \frac{1}{n} \begin{bmatrix} c_x k_{x_0} (1 - n\alpha_i \beta) & -\frac{c_x k_{x_0}}{2} (1 + n\alpha_i \beta) \\ -\frac{c_x k_{x_0}}{2} (1 + n\alpha_i \beta) & k_{x_0} (1 - n\alpha_i \beta) - c_x \end{bmatrix} \\ W_{R_i} &= \frac{1}{n} \\ &\times \begin{bmatrix} c_R k_{R_0} (1 - n\alpha_i \sigma_i) & -\frac{c_R}{2} (k_{\Omega_0} + B + n\alpha_i \sigma_i) \\ -\frac{c_R}{2} (k_{\Omega_0} + B + n\alpha_i \sigma_i) & k_{\Omega_0} (1 - n\alpha_i \sigma_i) - 2c_R \bar{\lambda} \end{bmatrix} \\ W_{q_i} &= \begin{bmatrix} c_q k_q & -\frac{c_q k_\omega}{2} \\ -\frac{c_q k_\omega}{2} & k_\omega - c_q \end{bmatrix} \\ W_{x_{R_i}} &= \alpha_i \begin{bmatrix} \gamma c_x k_{R_0} + \delta_i c_R k_{x_0} & \gamma c_x k_{\Omega_0} + \delta_i k_{x_0} \\ \gamma k_{R_0} + \delta_i c_R k_{x_0} & \gamma k_{\Omega_0} + \delta_i k_{x_0} \end{bmatrix} \\ W_{x_{q_i}} &= \begin{bmatrix} c_x B & 0 \\ \beta k_{x_0} e_{x_{\max}} + B & 0 \end{bmatrix} \quad W_{x_{R_i}} = \begin{bmatrix} c_R B & 0 \\ \alpha_0 \sigma_i k_{R_0} + B & 0 \end{bmatrix}. \end{aligned}$$

If the constants c_x , c_R , and c_q that are independent of the control input are sufficiently small, the matrices W_{x_i} , W_{R_i} , W_{q_i} are positive-definite. Also, if the error in the direction of the link is sufficiently small relative to the desired trajectory, we can choose the controller gains such that the matrix W_i is positive-definite, which follows that the zero equilibrium of tracking errors is stable in the sense of Lyapunov, and all of the tracking error variables z_i and the estimation error variables are uniformly bounded, i.e., e_{x_0} , $\dot{e}_{x_0}$, e_{R_0} , e_{Ω_0} , e_{q_i} , e_{ω_i} , $\tilde{\Delta}_{x_0}$, $\tilde{\Delta}_{R_0}$, $\tilde{\Delta}_{x_i} \in \mathcal{L}_\infty$. These also imply that e_{x_0} , $\dot{e}_{x_0}$, e_{R_0} , e_{Ω_0} , e_{q_i} , $e_{\omega_i} \in \mathcal{L}_2$ from (48), and that $\dot{e}_{x_0}$, $\ddot{e}_{x_0}$, $\dot{e}_{R_0}$, $\dot{e}_{\Omega_0}$, $\dot{e}_{q_i}$, $\dot{e}_{\omega_i} \in \mathcal{L}_\infty$. According to Barbalat's lemma [21], all of the tracking error variables e_{x_0} , $\dot{e}_{x_0}$, e_{R_0} , e_{Ω_0} , e_{q_i} , e_{ω_i} and their time-derivatives asymptotically converge to zero.

B. Proof of Proposition 2

We first show that the attitude of the i th quadrotor becomes exactly equal to its desired value within a finite time, i.e., $R_i(t) = R_{i_c}(t)$ for any $t \geq T$ for some $T > 0$. This is achieved by finite-time stability theory [18]. This proof is composed of two parts: 1) $s_i(t) = 0$ for any $t > T_s$ for some $T_s < \infty$ and 2) when the state is confined to the surface defined by $s_i = 0$, we have $e_{R_i}(t) = e_{\Omega_i}(t) = 0$ for any $t > T_R$ for some $T_R < \infty$. From now on, we drop the subscript i for simplicity, as the subsequent development is identical for all quadrotors.

From [9], the error dynamics for e_Ω is given by

$$J \dot{e}_\Omega = -\Omega \times \Omega + M + \Delta_R + J(\hat{\Omega} R^T R_c \Omega_c - R^T R_c \dot{\Omega}_c).$$

Substituting (36)

$$\begin{aligned} J \dot{e}_\Omega &= -k_s s - l_s S(r, s) + \Delta_R - B_\delta \frac{s}{\|s\|} \\ &\quad - (k_R J + l_s r J \text{diag}_j[|e_{R_j}|^{r-1}]) E(R, R_c) e_\Omega. \end{aligned} \quad (50)$$

Let a Lyapunov function be

$$\mathcal{W} = \frac{1}{2} s \cdot J s.$$

From (35) and (34), its time-derivative is given by

$$\dot{\mathcal{W}} = s \cdot \{J \dot{e}_\Omega + (k_R J + l_s r J \text{diag}_j[|e_{R_j}|^{r-1}]) E(R, R_c) e_\Omega\}.$$

Substituting (50) and (36), and using (15), it reduces to

$$\begin{aligned} \dot{\mathcal{W}} &\leq -k_s \|s\|^2 - l_s \sum_{j=1}^n |s_j|^{r+1} + B_\delta \|s\| - B_\delta \|s\| \\ &\leq -k_s \|s\|^2 - l_s \|s\|^{r+1} \end{aligned}$$

where the last inequality is obtained from the fact that $\|x\|^\alpha \leq \sum_{i=1}^n |x_i|^\alpha$ for any $x = [x_1, \dots, x_n]^T$ and $0 < \alpha < 2$ [20, Lemma 2]. Therefore

$$\dot{\mathcal{W}} \leq -\epsilon_1 \mathcal{W} - \epsilon_2 \mathcal{W}^{(r+1)/2}$$

where $\epsilon_1 = 2k_s/\lambda_M[J]$ and $\epsilon_2 = l_s(2/\lambda_M[J])^{(r+1)/2}$. This implies that $s(t) = 0$ for any $t \geq T_s$, where the settling time T_s satisfies

$$T_s \leq \frac{2}{\epsilon_1(1-r)} \ln \frac{\epsilon_1 \mathcal{W}(0)^{(1-r)/2} + \epsilon_2}{\epsilon_2}$$

according to [19, Remark 2].

Next, consider the second part of the proof when $s = 0$. Let a configuration error function for the attitude of a quadrotor be

$$\Psi_R = \frac{1}{2} \text{tr}[I - R_c^T R].$$

Consider a domain give by $D_R = \{(R, \Omega) \in \text{SO}(3) \times \mathbb{R}^3 \mid \Psi_R < \psi_R < 2\}$. It has been shown that the following inequality is satisfied in the domain:

$$\frac{1}{2} \|e_R\|^2 \leq \Psi_R \leq \frac{1}{2 - \psi_R} \|e_R\|^2. \quad (51)$$

Therefore, it is positive-definite about $e_R = 0$. The time-derivative of Ψ_R is given by $\dot{\Psi}_R = e_R \cdot e_\Omega$. Therefore, when $s = 0$, we have

$$\begin{aligned} \dot{\Psi}_R &= -k_R \|e_R\|^2 - l_R \sum_{j=1}^n |e_{R_j}|^{r+1} \\ &\leq -k_R \|e_R\|^2 - l_R \|e_R\|^{r+1}. \end{aligned}$$

Substituting (51), we obtain

$$\dot{\Psi}_R \leq -\epsilon_3 \Psi_R - \epsilon_4 \Psi_R^{(r+1)/2}$$

where $\epsilon_3 = k_R/2 - \psi_R$ and $\epsilon_4 = l_R/(2 - \psi_R)^{(r+1)/2}$. This implies that $e_R(t) = e_\Omega(t) = 0$ for any $t \geq T_R$, where the settling time T_R satisfies

$$T_R \leq \frac{2}{\epsilon_3(1-r)} \ln \frac{\epsilon_3 \Psi_R(0)^{(1-r)/2} + \epsilon_4}{\epsilon_4}.$$

In summary, whenever $t \geq T^* \triangleq \max\{T_s, T_R\}$, it is guaranteed that $R_i(t) = R_{i_c}(t)$ for the i th quadrotor. Next, we consider the *reduced system*, which corresponds to the dynamics of the payload and the rotational dynamics of the links when $R_i = R_{i_c}$. From (37) and (33), the control force of quadrotors when $R_i = R_{i_c}$ is given by

$$-f_i \cdot R_i e_3 = (u_i \cdot R_{c_i} e_3) R_{c_i} e_3 = \left(u_i \cdot \frac{u_i}{\|u_i\|} \right) - \frac{u_i}{\|u_i\|} = u_i.$$

Therefore, the reduced system is given by the controlled dynamics of the simplified model.

If the controller gains k_R , l_R , k_s , l_s are selected large such that T^* is sufficiently small, the solution stays inside of the

domain D given at (43) during $0 \leq t < T^*$. After $t \geq T^*$, the controlled system corresponds to the controlled system of the simplified dynamic model, and from Proposition 1, the tracking errors asymptotically coverage to zero, and the estimation error is uniformly bounded.

REFERENCES

- [1] L. S. Cicolani, G. Kanning, and R. Synnæstvedt, "Simulation of the dynamics of helicopter slung load systems," *J. Amer. Helicopter Soc.*, vol. 40, no. 4, pp. 44–61, 1995.
- [2] M. Bernard, "Generic slung load transportation system using small size helicopters," in *Proc. Int. Conf. Robot. Autom.*, May 2009, pp. 3258–3264.
- [3] I. Palunko, P. Cruz, and R. Fierro, "Agile load transportation: Safe and efficient load manipulation with aerial robots," *IEEE Robot. Autom. Mag.*, vol. 19, no. 3, pp. 69–79, Sep. 2012.
- [4] N. Michael, J. Fink, and V. Kumar, "Cooperative manipulation and transportation with aerial robots," *Auto. Robot.*, vol. 30, no. 1, pp. 73–86, 2011.
- [5] I. Maza, K. Kondak, M. Bernard, and A. Ollero, "Multi-UAV cooperation and control for load transportation and deployment," *J. Intell. Robot. Syst.*, vol. 57, pp. 417–449, Jan. 2010.
- [6] T. Lee, K. Sreenath, and V. Kumar, "Geometric control of cooperating multiple quadrotor UAVs with a suspended payload," in *Proc. IEEE Conf. Decision Control*, Dec. 2013, pp. 5510–5515.
- [7] F. Goodarzi, D. Lee, and T. Lee, "Geometric stabilization of a quadrotor UAV with a payload connected by flexible cable," in *Proc. Amer. Control Conf.*, 2014, pp. 4925–4930.
- [8] T. Lee, "Geometric control of multiple quadrotor UAVs transporting a cable-suspended rigid body," in *Proc. 53rd IEEE Conf. Decision Control*, Dec. 2014, pp. 6155–6160.
- [9] T. Lee, M. Leok, and N. McClamroch, "Geometric tracking control of a quadrotor UAV on SE(3)," in *Proc. 49th IEEE Conf. Decision Control*, Dec. 2010, pp. 5420–5425.
- [10] M. D. Shuster, "Survey of attitude representations," *J. Astron. Sci.*, vol. 41, no. 4, pp. 439–517, 1993.
- [11] T. Lee, "Computational geometric mechanics and control of rigid bodies," Ph.D. dissertation, Dept. Aerosp. Eng., Univ. Michigan, Ann Arbor, MI, USA, 2008.
- [12] T. Lee, M. Leok, and N. H. McClamroch, "Lagrangian mechanics and variational integrators on two-spheres," *Int. J. Numer. Methods Eng.*, vol. 79, no. 9, pp. 1147–1174, Aug. 2009.
- [13] T. Lee. (Mar. 2014). "Geometric control of multiple quadrotor UAVs transporting a cable-suspended rigid body." [Online]. Available: <https://arxiv.org/abs/1403.3684>
- [14] F. Goodarzi, D. Lee, and T. Lee, "Geometric nonlinear PID control of a quadrotor UAV on SE(3)," in *Proc. Eur. Control Conf.*, Jul. 2013, pp. 3845–3850.
- [15] F. Bullo and A. Lewis, *Geometric Control of Mechanical Systems*. Berlin, Germany: Springer-Verlag, 2005.
- [16] T.-H. Wu, "Spacecraft relative attitude formation tracking on SO(3) based on line-of-sight measurements," M.S. thesis, School Eng. Appl. Sci., George Washington Univ., Washington, DC, USA, 2012.
- [17] H. Khalil, *Nonlinear Systems*, 2nd ed. Englewood Cliffs, NJ, USA: Prentice-Hall, 1996.
- [18] S. P. Bhat and D. S. Bernstein, "Finite-time stability of continuous autonomous systems," *SIAM J. Control Optim.*, vol. 38, no. 3, pp. 751–766, Jan. 2000.
- [19] S. Yu, X. Yu, B. Shirinzadeh, and Z. Man, "Continuous finite-time control for robotic manipulators with terminal sliding mode," *Automatica*, vol. 41, no. 11, pp. 1957–1964, Nov. 2005.
- [20] S. Wu, G. Radice, Y. Gao, and Z. Sun, "Quaternion-based finite time control for spacecraft attitude tracking," *Acta Astron.*, vol. 69, nos. 1–2, pp. 48–58, Jul./Aug. 2011.
- [21] P. Ioannou and J. Sung, *Robust Adaptive Control*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1995.